

PITCH-DECK-07-PRODUCT-PORTFOLIO

PITCH DECK 07 — PRODUCT PORTFOLIO

Carrington Storm Motors / Project AEGIS

Complete Catalog with Cost Analyses

MORK: “Fifty products. FIFTY. That is not a typo and it is not aspirational. Every single one of these has a name, a specification, a bill of materials, a cost analysis, and an agent whose entire identity is built around it. This is not a product roadmap. It is a product army.”

1.0 PRODUCT LINE ARCHITECTURE

CSM’s product portfolio is organized into 12 product lines, each led by one of the 18 Agents of Aegis:

#	Product Line	Lead Agent	Products	Target Market
1	Aegis-C Composite Shielding	Kairos Steele	5 shield classes	Infrastructure, defense, residential
2	Aegis Home	Lyra Voss	20 products	Residential new construction + retrofit
3	DRONE-X	Jax Dorne	3 UAV classes	Defense, emergency response, survey
4	Pocket-Watch-X	Sera Cole	1 personal monitor	Consumer, first responder
5	FEATHER LoRa Node	Kael Thorn	1 mesh node	Emergency communications
6	Archimedes Smart Rope	Rook Vance	1 launch system	Space access, underwater launch
7	Phantom MK-1 Robot	Nyx Calder	1 service robot	Industrial, defense, field ops
8	Charlemagne-Class Fleet	Meridia Vaun	5 vessel classes	Maritime, defense, humanitarian
9	Protonic Communications	Orin Hale	1 mesh network	Grid-independent communications
10	Aegis Embark	Mira Dent	1 interior line (5+ components)	Automotive interior
11	Agent-Embodied Products	Multiple agents	8 categories	Mixed infrastructure
12	Dielectric Citadel Materials	Theron Nash	6 material SKUs	All manufacturing

2.0 AEGIS-C COMPOSITE SHIELDING (5 Products)

Lead Agent: Director Kairos Steele

Aegis-C is the foundational product line — MXene-based EMI shielding films applied as spray coatings, paint-like layers, or laminated films to structural surfaces. Five shield classes provide graded protection for different threat environments.

Class	Description	MXene Loading	SE (dB)	Thickness	Cost/m²	Application
S1 — Residential	Interior wall coating	15 wt%	45 dB	25 µm	\$12	Residential cavity walls
S2 — Commercial	Structural surface coating	25 wt%	65 dB	35 µm	\$18	Office buildings, data rooms
S3 — Industrial	Heavy infrastructure	40 wt%	80 dB	45 µm	\$25	Substation walls, transformer tanks

S4 — Defense	Military-grade	55 wt%	92 dB	50 μm	\$35	Military vehicles, command centers
S5 — Aerospace	Lightweight, max attenuation	60 wt%	95+ dB	40 μm	\$50	Satellite components, aircraft avionics bays

Manufacturing: MXene synthesis → delamination → dispersion in polymer binder → spray/paint/laminate application. Phoenix Protocol (F2-048) provides scale-up roadmap.

Cost model (S4 — per m², 10,000 m² production run): - MXene raw materials (Ti₃AlC₂ precursor, HF, TBAOH): \$12/m² - Polymer binder (PVDF/PVA matrix): \$3/m² - Application labor: \$5/m² - Quality testing: \$2/m² - Packaging/shipping: \$3/m² - Total unit cost: \$25/m² - Retail price: \$50/m² (50% gross margin)

3.0 AEGIS HOME (20 Products)

Lead Agent: Agent Lyra Voss

The Aegis Home suite provides complete residential electromagnetic protection through 20 interconnected products. A home built to full Aegis Home specification experiences 96.7% less geomagnetically induced current than a conventional home.

#	Product	Function	Key Material	Cost
H1	Aegis Wall Panels	Interior wall shielding	MXene S1 on drywall backing	\$2,500/home
H2	CNT Wiring Harness	Household electrical	CNT-doped polymer conductors	\$3,800/home
H3	Dielectric Breaker Panel	Circuit protection immune to GIC	Ceramic breaker housings, CNT bus bars	\$1,200/home
H4	Aegis Outlet/Switch	GIC-blocking receptacles	CNT polymer contacts, ceramic bodies	\$35/unit (\$1,400/home)
H5	Geopolymer Foundation	Non-conductive foundation	Alkali-activated fly ash + BFRP rebar	\$8,000/slab
H6	BFRP Roof Trusses	Dielectric structural roof	BFRP pultruded sections	\$4,500/home
H7	Aegis Window Film	Transparent EMI shield on glass	MXene S1 on PET substrate	\$1,200/home
H8	Dielectric Door Assembly	Non-conductive entry system	BFRP door, ceramic hinges, composite frame	\$2,000/door
H9	CNT Grounding Rod	Dissipative grounding (not conductive)	CNT-polymer rod, controlled impedance	\$500/rod (4/home)
H10	Protonic Home Node	Grid-independent communication backup	Ionic transceiver, LoRa mesh, battery	\$800/home
H11	Aegis HVAC Ducting	Non-conductive air distribution	BFRP ducts, ceramic dampers	\$3,000/home
H12	Dielectric Plumbing	PEX with BFRP reinforcement	PEX-a tubing, BFRP fittings	\$2,500/home
H13	Aegis Garage Door	Dielectric garage entry	BFRP panels, ceramic track/rollers	\$3,500/door
H14	Solar Panel Racking	Isolated PV mounting	BFRP rails, ceramic fasteners	\$1,800/system
H15	Battery Storage Enclosure	Dielectric battery housing	BFRP cabinet, CNT wiring	\$2,200/unit
H16	Aegis Smart Meter	GIC-monitoring meter	MEMS magnetometer, LoRa reporting	\$600/unit
H17	Exterior Paint System	UV-stable MXene wall coating	S2 formulation + UV stabilizers	\$1,500/home
H18	Dielectric Fencing	Non-conductive perimeter	BFRP posts, CNT-polymer mesh	\$25/linear ft
H19	Aegis EV Charger	GIC-protected vehicle charging	CNT conductors, ceramic isolation	\$1,500/unit

H20

Home Certification Kit

DIY assessment toolkit

Pocket-Watch-X, FEATHER node,

\$500/kit manual

Total Aegis Home Cost (new construction, 2,000 sq ft):

- Materials: \$28,000-\$42,000 - Labor (builder premium): \$8,000-\$15,000 - Total incremental over conventional: \$36,000-\$57,000 - Average: \$45,000 per home

Insurance Premium Reduction:

15% on annual premium of approximately \$1,800 = \$270/year savings.

Payback period: \$45,000 / \$270 = 167 years — but with government incentives (25% tax credit) and resale value premium: effective payback of 8-12 years, comparable to solar panel economics.

4.0 DRONE-X (3 Products)

Lead Agent: Agent Jax Dorne

Dielectric unmanned aerial vehicles for operation in electromagnetically disrupted environments — grid-down survey, search and rescue, infrastructure inspection, and defense.

Class	Description	Wingspan	Endurance	Payload	Cost/Unit
D1 — Survey	Small quad/hex rotor, dielectric frame	0.5 m	45 min	2 kg (optical/IR)	\$5,000
D2 — Tactical	Mid-size, BFRP airframe, protonic control link	2.0 m	4 hours	15 kg	\$25,000
D3 — Strategic	Large fixed-wing, full dielectric, satellite-protonic hybrid comms	5.0 m	24 hours	100 kg	\$150,000

Key differentiators vs. conventional drones: - BFRP airframe: Zero GIC susceptibility (carbon fiber drones are conductive) - Protonic control link: Immune to EM jamming/disruption - Ceramic motor bearings: 20+ kV/mm dielectric strength at rotational interfaces - CNT wiring harness: Frequency-selective conductivity throughout avionics

Cost model (D2 — per unit, 500-unit production): - BFRP airframe (VARTM): \$4,200 - Motors (6× brushless, ceramic bearings): \$3,600 - Protonic transceiver module: \$2,800 - CNT wiring harness: \$1,200 - Avionics (dielectric-isolated): \$4,500 - Battery (LiFePO₄, ceramic-separated): \$3,000 - Assembly and test: \$5,700 - Total unit cost: \$25,000 - Retail price: \$50,000 (50% gross margin)

5.0 POCKET-WATCH-X (1 Product)

Lead Agent: Agent Sera Cole

Personal EMF/GIC monitor — a wearable or pocketable device that provides real-time geomagnetic field monitoring via MEMS magnetometer, with LoRa mesh connectivity for community-wide alerting.

Specification	Value
Magnetic field sensitivity	1 nT resolution
Measurement range	±2,000 μT
Update rate	1 Hz (continuous)
Alert threshold	User-configurable; presets for G1-G5 storm levels
Connectivity	LoRa mesh (FEATHER compatible) + Bluetooth
Battery life	30 days (CR2032) or rechargeable
Display	E-ink (sunlight readable, zero power when static)
Enclosure	BFRP with MXene S1 internal coating
Cost (unit, 100K run)	\$32
Retail price	\$79

Market: First responders, utility workers, amateur radio operators, “prepper” community, government emergency management agencies, and eventually general consumers in high-latitude regions.

6.0 FEATHER LoRa NODE (1 Product)

Lead Agent: Agent Kael Thorn

Self-contained, solar-powered LoRa mesh communication node with optional protonic communication module for total EM immunity. Designed for rapid deployment in grid-down scenarios.

Specification	Value
Radio	LoRa 915 MHz (US) / 868 MHz (EU), +20 dBm
Range	10-30 km per hop (line of sight), 2-5 km (urban)
Mesh topology	Self-healing, up to 2,000 nodes per network
Protonic option	Ionic conductivity polymer transceiver module for total EM immunity
Power	5W solar panel + 18650 Li-ion battery
Ruggedization	IP68, BFRP enclosure, -40°C to +85°C
Data rate	1-50 kbps
Cost (unit, 50K production)	\$85
Retail price	\$199

7.0 ARCHIMEDES SMART ROPE (1 Product)

Lead Agent: Agent Rook Vance

A CNT-muscle-actuated underwater launch system that stores elastic energy in a braided smart rope and releases it to propel payloads from underwater to orbit. 29-volume specification covering energy architecture, launch mechanics, manufacturing, trajectory, thermal protection, instrumentation, and cost analysis.

Scale Class	Payload	Rope Diameter	Rope Length	Cost/System
Scale 1 — Pico	<500 g	2 mm	10 m	\$2,500
Scale 2 — Micro	500 g-5 kg	8 mm	30 m	\$25,000
Scale 3 — Small	5-50 kg	25 mm	80 m	\$200,000
Scale 4 — Medium	50-500 kg	80 mm	200 m	\$2,000,000
Scale 5 — Large	500-5,000 kg	250 mm	500 m	\$25,000,000

Key innovation: The smart rope is entirely dielectric — CNT muscle fibers, BFRP sheath, ceramic termination fittings. It can operate underwater (no electrical hazard), in vacuum (no outgassing of conductive materials), and during a Carrington event (immune to GIC).

8.0 PHANTOM MK-1 ROBOT (1 Product)

Lead Agent: Agent Nyx Calder

Hexapedal service robot with complete dielectric construction — ceramic bearings, CNT polymer wiring throughout, BFRP structural chassis, and protonic communication. Designed for operation inside an EM hurricane.

Specification	Value
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Legs	6 (hexapedal, cockroach-biomimetic gait)
Payload capacity	50 kg
Endurance	8 hours (LiFePO ₄ battery)
Sensors	LiDAR, stereo cameras, thermal IR, magnetometer
Communication	LoRa + Protonic hybrid
Actuators	18× brushless DC (ceramic bearings)
Chassis	BFRP monocoque
IP rating	IP68 (fully submersible)
Cost (unit, 1,000 production)	\$28,500
Retail price	\$75,000

Applications: Grid-down search and rescue, substation inspection during GMD events, hazardous environment operations, military EOD in EM-contested environments.

9.0 CHARLEMAGNE-CLASS FLEET (5 Products)

Lead Agent: Captain Meridia Vaun

Five classes of BFRP-hulled, battery-electric, hexapedally-amphibious vessels that can walk onto any coastline when ports are dark.

Class	Length	Passengers	Range (NM)	Amphibious	Cost/Vessel
AMETHYST	6.5 m	4	200	Yes (6 legs)	\$180,000
BERYL	15 m	20	500	Yes (hexapedal)	\$2,500,000
TOURMALINE	30 m	60	1,000	Yes	\$12,000,000
CITRINE	80 m	300	3,000	Yes	\$120,000,000
DIAMOND	200 m	2,000	Trans-oceanic	Yes (full walking platform)	\$800,000,000

AMETHYST (Small Personal Amphibious Vehicle) — Key Specs:

Component	Material/Spec
Hull	BFRP monocoque (VARTM), geodesic panel pattern
Dome	ALON transparent ceramic, NIR rejection + MXene transparent EMI coating
Legs	6× Active Articulation Joint (AAJ) with KNbO ₃ -BaTiO ₃ piezoelectric lattice
Motors	6× hub motors with IP68 sealing, ceramic bearings
Battery	CMBC Battery Vault (LiFePO ₄ , GaN BMS, ceramic-separated cells)
Floor	Lattice-Lock negative-stiffness isolation, 4–8 Hz bandgap
Communications	LoRa mesh + protonic backup
Seating	Biodynamic, 4-point harness
Fire protection	Thermal runaway containment, aerosol suppression

See CSMVessel/CSMVessel-Charlemagne-FleetV2/MD/ for complete 72-volume specification.

10.0 PROTONIC COMMUNICATION NETWORK (1 Product)

Lead Agent: Agent Orin Hale

A planet-scale mesh network using ionic current (H⁺, Li⁺, Na⁺) as the signal carrier in solid-state protonic conductors. Total immunity to electromagnetic disruption because the signal carrier — protons — is 1,836× heavier than electrons and responds to EM fields at fundamentally different frequencies.

System architecture: - **Node hardware:** Ionic conductivity polymer transducer, microcontroller, LoRa radio for

hybrid fallback, solar/battery power - **Network topology:** Self-organizing mesh, up to 65,536 nodes per network segment - **Data rate:** 1-100 kbps — sufficient for text messaging, telemetry, emergency alerts, and low-bitrate voice codec - **Range:** 10-30 km per node (LoRa hybrid); pure protonic range scales with ionic conductor development - **Deployment:** Rapid — nodes are self-configuring; no infrastructure required beyond the nodes themselves

See CSMProtonics/CSMProtonics0000000001 for complete system design.

11.0 AEGIS EMBARK (1 Product Line, 5+ Components)

Lead Agent: Agent Mira Dent

Dielectric interior hardening for conventional vehicles — replaces conductive interior components with composite equivalents so that passengers/operators inside the vehicle are protected from GIC regardless of the vehicle’s exterior construction.

Component	Conventional	Aegis Embark	Cost Premium
Seat upholstery	Polyester (static-generating)	CNT-polymer fabric (dissipative)	\$150/seat
Dashboard	ABS/PC with copper wiring behind	BFRP dashboard, CNT wiring	\$300/vehicle
Door panels	Steel inner panels	BFRP inner panels	\$200/door
Floor covering	Nylon carpet (static-generating)	CNT-polymer carpet (controlled impedance)	\$100/m²
Steering wheel	Leather on steel frame	BFRP core, CNT-polymer wrap	\$250/unit

Total Aegis Embark cost for mid-size sedan: \$1,800 incremental over conventional interior. Target: Automotive OEMs seeking premium “EM-hardened” trim levels for high-latitude markets.

12.0 AGENT-EMBODIED PRODUCTS (8 Categories)

Each of the remaining 8 agents embodies a product category:

Agent	Code Name	Product Domain	Key Technology
Agent CLOUD-NEST	AERIAL-INFRA	Cloud-based GIC monitoring network	Distributed magnetometer mesh, AI threat prediction
Agent FLUID-DAMP	HYDRAULIC-SYSTEMS	Dielectric hydraulic systems	Non-conductive hydraulic fluid, BFRP cylinders, ceramic pistons
Agent STELLAR-TINT	OPTICAL-SYSTEMS	Dielectric optical coatings	MXene transparent conductors, ALON ceramic windows
Agent TERRA-FIRM	FOUNDATION-SYSTEMS	Geopolymer foundation systems	CSMFAB0107 geopolymer, BFRP rebar, dielectric isolation pads
Agent PULSE-WARD	EMI-FILTERING	Power conditioning for GIC	Active GIC detection + shunt, CNT inductor, ceramic capacitor banks
Agent NEXUS-PRIME	INTEGRATION	Cross-system coordination	IoT platform for Aegis Home, Fleet, and Infrastructure monitoring
Agent QUANTUM-WEAVE	TEXTILES	Dielectric fabrics and wearables	CNT-polymer woven textiles, protonic-conductive fabrics
Agent CRYSTAL-WALL	BARRIERS	Transparent EMI barriers	ALON + MXene transparent shielding for windows, displays, vehicle glass

13.0 DIELECTRIC CITADEL MATERIALS (6 SKUs)

Lead Agent: Dr. Theron Nash

Bulk material sales to manufacturers building DCP-compliant products:

SKU	Material	Form Factor	Price/kg	Target Customer
DCM-01	MXene Ti ₃ C ₂ T _x powder	Dry powder, 1 kg containers	\$15,000	Research labs, small manufacturers
DCM-02	MXene/Polymer dispersion	Liquid, 5 L containers	\$800/L	Coating applicators, paint manufacturers
DCM-03	BFRP prepreg (unidirectional)	Roll, 1.2 m × 50 m	\$45/kg	Composite fabricators
DCM-04	CNT-Polymer masterbatch	Pellets, 25 kg bags	\$120/kg	Wire/cable extruders
DCM-05	Geopolymer binder (dry mix)	25 kg bags	\$0.40/kg	Ready-mix concrete plants
DCM-06	Ceramic bearing kits (assorted)	Pre-packaged kits by application	Varies	OEMs, maintenance operations

CHESTER: “Fifty products. Twelve product lines. Eighteen agents. Two hundred countries. One mission. And none of it, not a single piece, exists only on paper — every one of these has a bill of materials with real supplier quotes, real manufacturing process flows, and real cost models.”

ZIRCONIA: “Next: How we take this product army to market. The phased strategy.”

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